Rapid Field report – Nepal Airborne Gravity Survey

Arne Vestergaard Olesen / Rene Forsberg, DTU-Space, Dec 17, 2010

The Nepal airborne gravity survey was initiated on Nov 26, with the arrival of Arne Vestergaard Olesen and Rene Forsberg from Denmark. Aircraft OY-CKP, owned and operated by COWI, arrived the day after. The field team from Denmark consisted of

Arne V. Olesen – field leader
Rene Forsberg – project leader
Indridi Einarson – post.doc. (newly employed)
Christian Hoeg Andersen – chief pilot
Morten Rasmussen - pilot
Anders Halvgaard - pilot (replaced Morten on Nov 15)

The crew was lodged at Hotel Everest. RF left Nepal in the period Nov 30-Dec 9 for meetings in Denmark. The aircraft operations base was the Nepal Army Air Wing base at Tribhuvan Airport.

From the Department of Survey, several from the staff participated in field operations, logistics and GPS-support, as well as military liason. The PI's for arrangements was *Niraj Manandhar* and *Khalyan Shresta.*

The first flight day was on Dec 4, following training course at the Nepal Civil Aviation Authority, and a total of 13 survey flights was carried out, on 12 days. A failure in the aircraft power system on Dec 13 gave a 2-day waiting period for spare parts brought in from Denmark.

Generally the flight conditions were excellent in the lower areas, but very challenging in the mountains, with extremely strong winds due to the jet stream (airspeed on survey lines varied from 130 to 380 knots, depending on direction of flight at the high altitudes). Flight levels up to 31000 ft were flown. The table below shows the flight times. The line numbers are shown in Figure 1.

Nepal 2010: OY-CKP flights (time is UTC)

Date/JD	Lines		Airborne	Landed	Airborne time	Airborne time to date
04 Dec / 338	M2G1, N1N2, G2G3, H3H2		0358	0845	4h47	
05 Dec / 339	U1U2, T2T1		0909	1128	2h19	7h06
06 Dec / 340	K2K1, L1L2, L2L3, K3K2		0236	0857	6h21	13h27
07 Dec / 341	R2R1, Q1Q2, P2P1, O1O2		0249	0850	6h01	19h28
08 Dec / 342	J2J3, I3I2,		0331	0545	2h14	21h42
---	J2J1, I1I2		0755	1134	3h39	25h21
09 Dec / 343	H2H1, G1G2, F2F3, E3E2		0514	1106	5h52	31h13
10 Dec / 344	E2E1, F1F2, C2C3, D3D2		0323	0841	5h18	36h31
11 Dec / 345	C2C1, D1D2, A2A1, X5X6		0256	0751	4h55	41h26
12 Dec / 346	X3X4, S2S1, X2X1, B1B2		0237	0751	5h14	46h40
13 Dec / 347	V1V2		0334	0606	2h32	49h12
16 Dec / 350	O2O1, T3T4, X13X14, K5K56		0500	0956	4h56	54h08
17 Dec / 351	X10X9, Y1Y2, X7X8		0340	0604	2h24	56h32
Total 56h 32min						

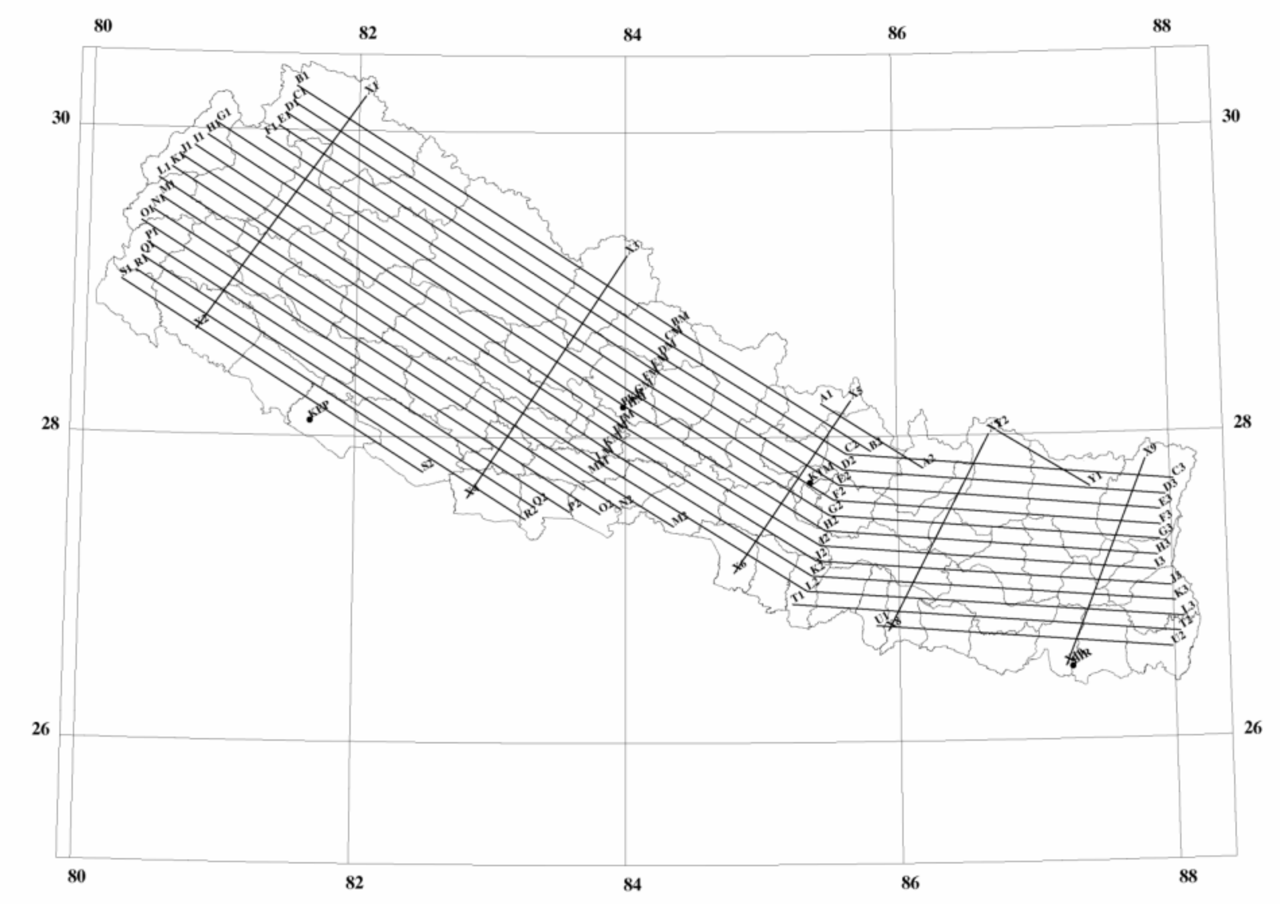


Fig. 1. Planned and flown flight tracks. Additional lines were added to the SW and SE.

This very brief field report will be updated when the preliminary GPS has been processed.

A preliminary geoid report has been done based on data up to December 16.

Initial training and delivery of equipment (first PC) to "Geodetic Lab" was done on Dec 14-15.

/RF

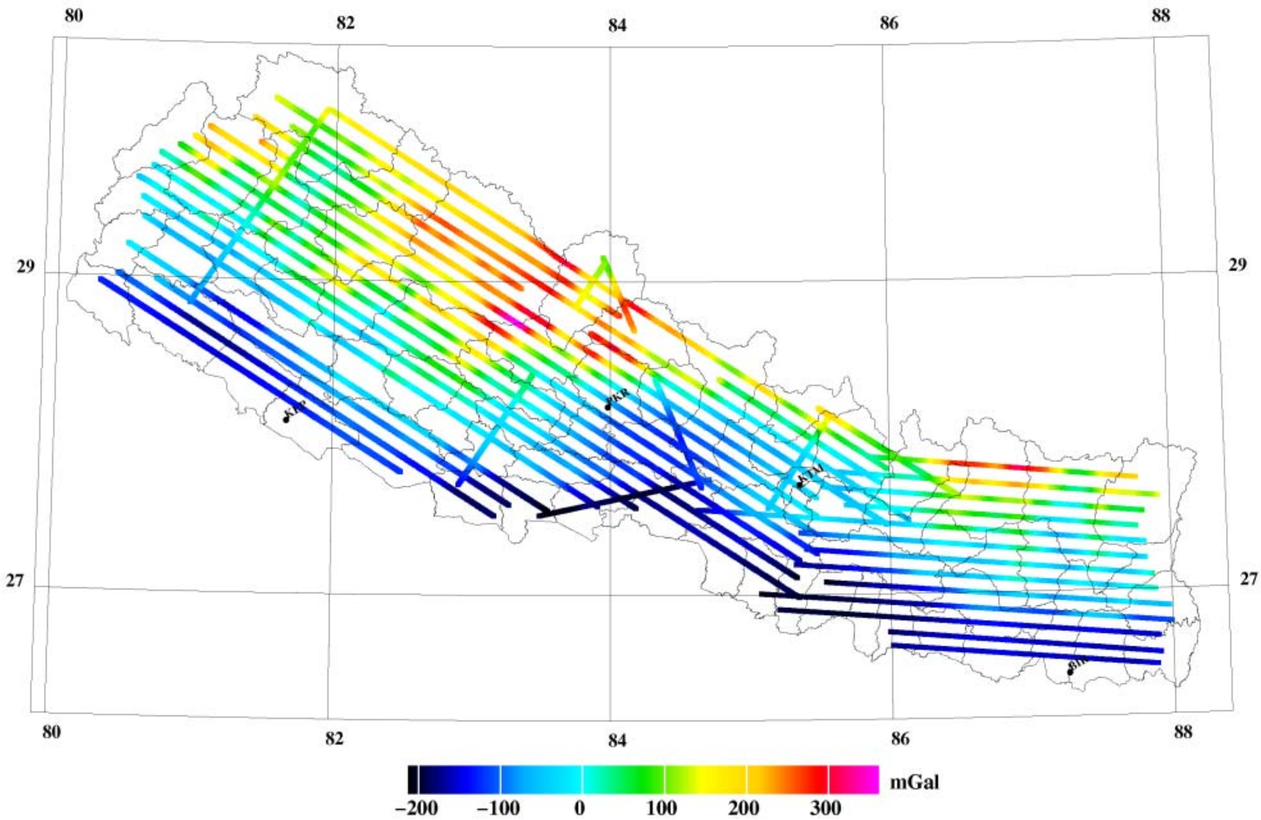


Fig. 2. Free-air anomalies at altitude (last 2 flights not shown)

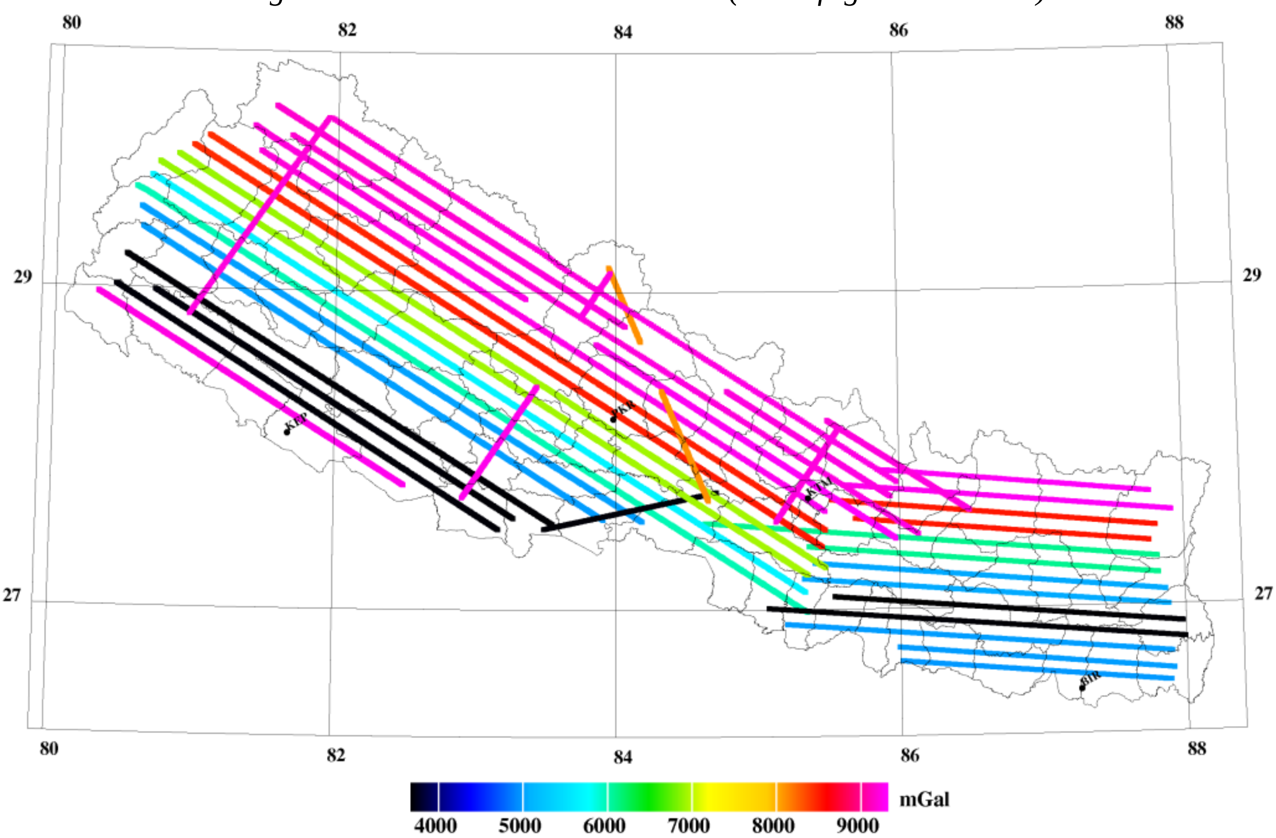
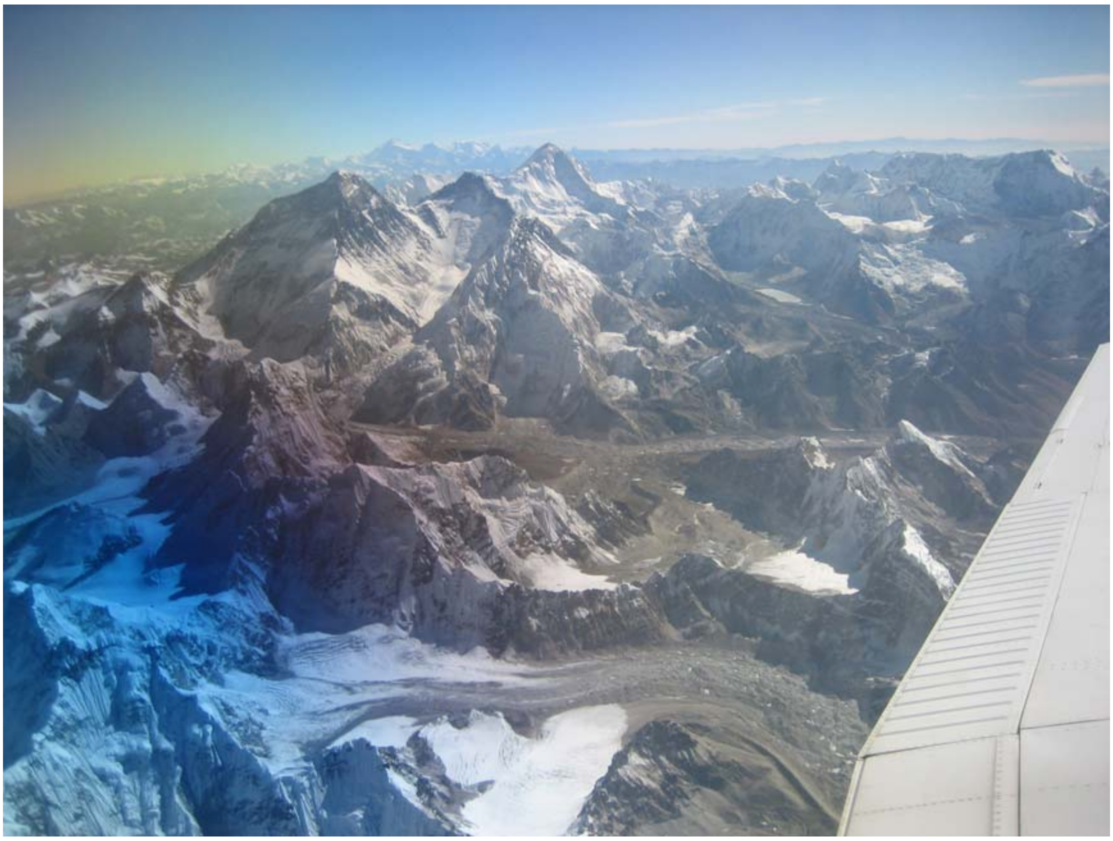


Fig. 3. Height of flight tracks in meter

COWI OY-CKP Beech King Air and flight crew (last day)



Mt Everest and Makalu from the beginning of line X7-X8